

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Software Engineering Mini-Project

Software Test Plan

Online Internship and Placement Preparation Portal

Project: Online Internship and Placement Preparation Portal

Group: Team 16

Version: 1.0

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Status: Draft for Review

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Revision History

Version	Date	Author	Change Summary
1.0	06-10-2026	Rahul Bellary	Part1- se submission

1. Introduction

Purpose: This document defines the test plan for the Online Internship and Placement Preparation Portal v1.0. It describes objectives, scope, strategy, environment, schedule and responsibilities for testing.

Scope: Testing covers authentication, internship listings, applications, aptitude tests, progress analytics, administration and the non-functional and security requirements in the SRS. Third-party services (SMTP provider, cloud infrastructure) are excluded.

References: SRS v1.0, SAD v1.0, OWASP ASVS 4.0, IEEE Std 829-2008.

Definitions: STP (Software Test Plan), SRS, SAD, RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), RBAC.

2. Test Items

- Auth & User module (registration, login, profile)
- Internship module (listing, moderation, search)
- Application module (apply, status, notification)
- Aptitude Test module (question bank, attempt, grading)
- Analytics service and dashboards (Python)
- Admin module (users, reports, audit log)
- React web client

3. Features to be Tested

Features are mapped to SRS requirement IDs.

Req. ID	Feature	Type
PIP-F-001	The system shall allow a visitor to register as a Student or Recruiter by providing name, email and a password of at least 8 characters containing at least one letter and one digit	Functional
PIP-F-002	The system shall authenticate users using email and password and issue a session token valid for 60 minutes	Functional

PIP-F-003	The system shall allow a Student to create and edit a profile containing education, skills, and a resume in PDF format of at most 5 MB	Functional
PIP-F-004	The system shall allow a Recruiter to create an internship listing with title, company, description, required skills, stipend, location, mode (remote/onsite/hybrid) and application deadline	Functional
PIP-F-005	The system shall allow an Admin to approve or reject a Pending listing with an optional remark	Functional
PIP-F-006	The system shall allow a Recruiter to edit or close only the listings created by that Recruiter	Functional
PIP-F-007	The system shall allow a Student to search Approved listings by keyword and filter by skill, location, mode and stipend range	Functional
PIP-F-008	The system shall allow a Student to apply to an Approved listing before its deadline using the resume in the student profile	Functional
PIP-F-009	The system shall allow a Student to view all own applications with the current status: Submitted, Under Review, Shortlisted, Rejected or Selected.	Functional
PIP-F-010	The system shall allow a Recruiter to view applicants (profile and resume) of own listings and to update the application status.	Functional
PIP-F-011	The system shall notify a Student through an in-app notification and an email within 1 minute of any change in the status of the application.	Functional
PIP-F-012	The system shall allow an Admin to create aptitude tests and manage a question bank	Functional
PIP-F-013	The system shall allow a Student to start a timed aptitude test	Functional
PIP-F-014	The system shall auto-grade a submitted attempt and display the total score and the category-wise score to the Student immediately after submission.	Functional
PIP-F-015	The system shall allow a Student to review past attempts, including the correct answers, only after the attempt has been submitted	Functional
PIP-F-016	The system shall save the selected answers of an active attempt at regular intervals so that the Student can resume the attempt after a connection loss within the remaining test time.	Functional
PIP-F-017	The system shall display a Student dashboard showing applications by status, the trend of test scores across attempts, and category-wise performance.	Functional
PIP-F-018	The system shall recommend practice categories to a Student based on the lowest category-wise scores of the Student, computed by the analytics service.	Functional
PIP-F-019	The system shall display aggregate analytics to an Admin: applications per listing, average test scores per category, and the number of Selected students.	Functional
PIP-F-020	The system shall allow an Admin to export registration and application reports in CSV format.	Functional
PIP-F-021	The system shall allow an Admin to deactivate and reactivate user accounts	Functional
PIP-F-022	The system shall record an audit log entry (actor, action, target, timestamp) for listing approvals, application status changes and account deactivations.	Functional
PIP-NF-001	At least 95% of API requests (excluding analytics and report export) shall complete within 2 seconds with 500 concurrent users.	Non-functional (Performance)
PIP-NF-002	A submitted test attempt shall be graded and the result returned within 3 seconds for a test of up to 50 questions.	Non-functional (Performance)
PIP-NF-003	The portal shall be available at least 99.5% of the time in a calendar month, excluding planned maintenance announced 24 hours in advance.	Non-functional (Availability)

PIP-NF-004	A first-time Student shall be able to register, complete the profile and submit a first application within 10 minutes without training. The UI shall be responsive from 360 px screen width.	Non-functional (Usability)
PIP-NF-005	The web application shall work on the latest two stable versions of Chrome, Firefox and Edge.	Non-functional (Compatibility)
PIP-NF-006	The system shall support 5,000 registered users and 2,000 test attempts per day, and the application tier shall be horizontally scalable.	Non-functional (Scalability)
PIP-NF-007	Backend modules shall have at least 70% unit-test line coverage and shall pass the ESLint (JavaScript) and flake8 (Python) checks without errors.	Non-functional (Maintainability)
PIP-NF-008	The database shall be backed up daily and restorable with a data loss of at most 24 hours.	Non-functional (Reliability)
PIP-S-001	All client-server and service-to-service communication shall use TLS 1.2 or higher.	Security
PIP-S-002	Passwords shall be stored only as bcrypt hashes with cost factor 10 or higher; plain-text passwords shall never be stored or logged.	Security
PIP-S-003	The system shall enforce role-based access control (Student, Recruiter, Admin) on every API endpoint. A request outside the role's permission shall return HTTP 403.	Security
PIP-S-004	All user input shall be validated on the server and sanitised to prevent NoSQL injection and cross-site scripting. Invalid input shall return HTTP 400.	Security
PIP-S-005	The system shall limit requests to 100 per minute per IP address and shall return HTTP 429 when exceeded.	Security
PIP-S-006	A resume shall be accessible only to its owner, Admins, and Recruiters of the listing to which the Student applied.	Security
PIP-S-007	The system shall write security events (login success/failure, logout, role-violation attempts) to an append-only audit log without storing passwords or tokens.	Security

4. Features Not to be Tested

- Internal behaviour of the SMTP provider (only the integration is tested).
- MongoDB and Node.js framework internals (assumed tested by vendors).
- Cloud hosting infrastructure reliability beyond the measured availability checks.
- Browsers other than those listed in PIP-NF-005.
- Internationalisation (English only in v1.0).

5. Test Approach / Strategy

Levels:

- Unit tests: Jest (Node.js modules, React components), pytest (analytics service).
- Integration tests: API-to-database, API-to-analytics service and API-to-SMTP stub using Supertest.
- System tests: end-to-end flows on the deployed build using Cypress.

- Acceptance tests: UAT with sample Student, Recruiter and Admin users.

Types:

- Functional testing of all features.
- Regression testing on every merged pull request (automated suite).
- Performance testing (response time and load) with JMeter.
- Usability and compatibility testing (browsers, 360 px viewport).
- Security testing (see 5.1).

Entry Criteria: stable build deployed to the test environment, test data loaded, test cases reviewed.

Exit Criteria: 100% of planned test cases executed, no open critical or high defects, at least 90% of test cases passed, all acceptance criteria satisfied.

5.1 Security Validation

Area	Validation	Requirement	Test Case
Authentication	Brute-force lockout, token expiry, no user enumeration in error messages	PIP-F-002	TC-AUTH-03
Authorisation	RBAC on every endpoint; resume access limited to owner/Admin/applied Recruiter	PIP-S-003, PIP-S-006	TC-SEC-01
Input handling	NoSQL injection, XSS, oversized and invalid files; fuzzing of login, profile, search and file-upload inputs	PIP-S-004	TC-SEC-02
Transport and storage	TLS 1.2+ verification, bcrypt hash verification, no secrets or passwords in logs	PIP-S-001, PIP-S-002, PIP-S-007	TC-SEC-03
Availability	Rate limiting returns HTTP 429	PIP-S-005	TC-SEC-03
Dependency and scan	npm audit / pip-audit and OWASP ZAP baseline scan on the deployed build	All security	Scan report

5.2 Test Cases

The following test cases cover the functional, non-functional and security requirements.

ID	Type	Req. ID	Title	Steps	Expected Result
TC-AUTH-01	Functional	PIP-F-001	Register with valid details	1. Open Register page. 2. Enter valid name, new email, valid password, role Student. 3. Submit.	Account created; user is redirected to login with a success message.
TC-AUTH-02	Functional (negative)	PIP-F-001	Register with duplicate email and weak password	1. Register using an already registered email. 2. Register a new email with a password lacking a digit.	Both attempts rejected with specific error messages; no account created.

TC-AUTH-03	Functional	PIP-F-002	Login lockout after repeated failures	1. Enter wrong password 5 times for a valid account. 2. Enter the correct password immediately.	Account locked; correct password is refused until the lockout period ends; lock message displayed.
TC-INT-01	Functional	PIP-F-004, PIP-F-005	Listing approval workflow	1. Recruiter creates a listing. 2. Student searches for it. 3. Admin approves. 4. Student searches again.	Listing is Pending and not visible to the Student at step 2; visible after approval at step 4.
TC-INT-02	Functional	PIP-F-007	Search and filter listings	1. Login as Student. 2. Search by keyword and apply skill, location and stipend filters.	Only matching Approved listings are returned, paginated at 10 per page.
TC-APP-01	Functional	PIP-F-008, PIP-F-009	Apply for an internship	1. Student with uploaded resume opens an Approved listing. 2. Click Apply. 3. Open My Applications.	Application created with status Submitted and listed under My Applications.
TC-APP-02	Functional (negative)	PIP-F-008	Duplicate and late application	1. Apply again to the same listing. 2. Apply to a listing whose deadline has passed.	Both requests rejected with clear error messages; no new application stored.
TC-APP-03	Functional	PIP-F-010, PIP-F-011	Status update and notification	1. Recruiter changes an application to Shortlisted. 2. Check Student notifications and email.	Status updated; in-app notification and email received within 1 minute.
TC-TST-01	Functional	PIP-F-013, PIP-F-016	Timed test auto-submit and resume	1. Start a test. 2. Answer some questions and disconnect, then reconnect before time ends. 3. Let the timer expire.	Saved answers restored on reconnect; attempt is auto-submitted at timeout.
TC-TST-02	Functional	PIP-F-014, PIP-F-015	Grading and answer visibility	1. Submit a completed attempt. 2. Inspect the question API response during an active attempt. 3. Open the attempt review after submission.	Total and category scores shown immediately; correct answers absent from active-attempt responses; shown only in the review.
TC-ANL-01	Functional	PIP-F-017, PIP-F-018	Dashboard and recommendations	1. Complete two attempts with different category scores. 2. Open the dashboard.	Dashboard shows application status counts, score trend, category chart, and recommends the weakest category.
TC-PERF-01	Non-functional (performance)	PIP-NF-001	Load test with 500 concurrent users	1. Use JMeter to simulate 500 users running search, login and dashboard flows for 10 minutes.	At least 95% of requests complete within 2 seconds; error rate stays within the agreed limit.
TC-PERF-02	Non-functional (performance)	PIP-NF-002	Grading response time	1. Submit a 50-question attempt. 2. Measure time to the result response (repeat 20 times).	Result returned within 3 seconds in all measured runs.
TC-USA-01	Non-functional (usability)	PIP-NF-004, PIP-NF-005	Usability and browser compatibility	1. Five new users register and apply without guidance, timed. 2. Repeat key flows on Chrome, Firefox, Edge and on a 360 px viewport.	Users finish within 10 minutes; layout and flows work on all listed browsers and the small viewport.
TC-SEC-01	Security	PIP-S-003, PIP-S-006	Role-based access control	1. Login as Student and call an Admin endpoint and another student's resume URL. 2. Login as Recruiter and request resume of a non-applicant.	All requests return 403; no data is exposed; the attempts are logged.
TC-SEC-02	Security	PIP-S-004	Injection and XSS handling	1. Submit NoSQL operators in the login body. 2. Enter a script tag in profile and listing text fields. 3. View the rendered pages.	Injection rejected with 400; script content is escaped and never executed.

TC-SEC-03	Security	PIP-S-001, PIP-S-002, PIP-S-005	Transport, password storage and rate limiting	1. Access the site over HTTP and inspect the TLS version. 2. Inspect stored password values. 3. Send requests above the rate limit.	HTTP redirected to HTTPS with TLS 1.2+; only bcrypt hashes stored; HTTP 429 after the limit.
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6. Test Environment

- **Hardware:** developer laptops and one cloud VM (2 vCPU, 4 GB RAM) for the staging deployment.
- **Software:** Portal v1.0 (React, Node.js 20, Python 3.11 service, MongoDB 7), Chrome, Firefox, Edge.
- **Tools:** Jest, pytest, Supertest, Cypress, Postman (API), JMeter (performance), OWASP ZAP (security), GitHub Issues (defect tracking).
- **Test data:** seeded Student, Recruiter and Admin accounts, sample listings, a question bank covering all four categories.

7. Test Schedule

Milestone	Date
Test plan and test case design	05-Oct-2026
Environment setup and test data	12-Oct-2026
Unit and integration testing	13-Oct-2026 to 27-Oct-2026
System, performance and security testing	28-Oct-2026 to 06-Nov-2026
UAT	09-Nov-2026 to 11-Nov-2026
Test summary report	13-Nov-2026

8. Test Deliverables

- Test Plan (this document)
- Test cases (manual and automated) and test scripts
- Test data
- Test execution logs
- Defect reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
Test Lead	Rahul Bellary	Prepare plan, coordinate execution, summary report
Test Engineer	<Team Member 2 Name>	Design and execute test cases, log defects
Test Engineer	<Team Member 3 Name>	Automation, performance and security tests

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build	Request early smoke builds; test by module as they complete
Staging environment downtime	Keep a local Docker Compose copy of the environment
SMTP provider limits during notification tests	Use an SMTP stub/mail trap for tests
Insufficient question bank data	Prepare a seed script with sample questions
Small team size	Automate regression tests; prioritise High-priority requirements

11. Assumptions & Dependencies

- Staging environment and seed data are available before system testing.
- SMTP mail trap is configured.
- Requirement IDs in SRS v1.0 remain stable.

12. Suspension & Resumption Criteria

Suspend testing if: the environment is unavailable for more than 4 hours; or the build is so unstable that more than 30% of test cases are blocked.

Resume testing when: blocking defects are fixed and verified; the environment is stable again.

13. Test Case Management & Traceability

The RTM maps SRS requirements to test cases. Test cases are tracked in a shared spreadsheet and results updated per execution cycle.

Requirement ID	Test Case IDs
PIP-F-001	TC-AUTH-01, TC-AUTH-02

PIP-F-002	TC-AUTH-03
PIP-F-004	TC-INT-01
PIP-F-005	TC-INT-01
PIP-F-007	TC-INT-02
PIP-F-008	TC-APP-01, TC-APP-02
PIP-F-009	TC-APP-01
PIP-F-010	TC-APP-03
PIP-F-011	TC-APP-03
PIP-F-013	TC-TST-01
PIP-F-014	TC-TST-02
PIP-F-015	TC-TST-02
PIP-F-016	TC-TST-01
PIP-F-017	TC-ANL-01
PIP-F-018	TC-ANL-01
PIP-NF-001	TC-PERF-01
PIP-NF-002	TC-PERF-02
PIP-NF-004	TC-USA-01
PIP-NF-005	TC-USA-01
PIP-S-001	TC-SEC-03
PIP-S-002	TC-SEC-03
PIP-S-003	TC-SEC-01
PIP-S-004	TC-SEC-02
PIP-S-005	TC-SEC-03
PIP-S-006	TC-SEC-01

Requirements not listed above are covered by unit and integration tests and will be added to the RTM as test cases are expanded.

14. Test Metrics & Reporting

- % of test cases executed
- % passed / failed
- Defect density and defect aging
- Requirement coverage
- Daily execution status and final Test Summary Report

Approvals

Role	Name	Signature / Date
Team Member	Rahul Bellary	
Team Member	Dhoni Satwik	
Team Member	Prateek Hachadad	
Course Instructor		